

Credit Risk Analytics

DataDNA
Challenge
September
2025

Overview

Calculator

Simulation



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Loan Portfolio
104.8M



Borrowers
10.9K

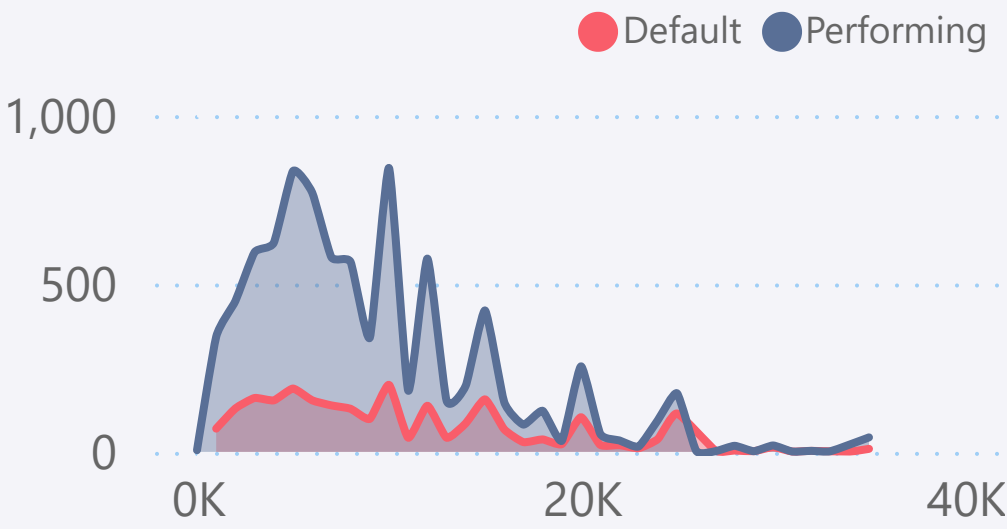


Default Rate
21.7%

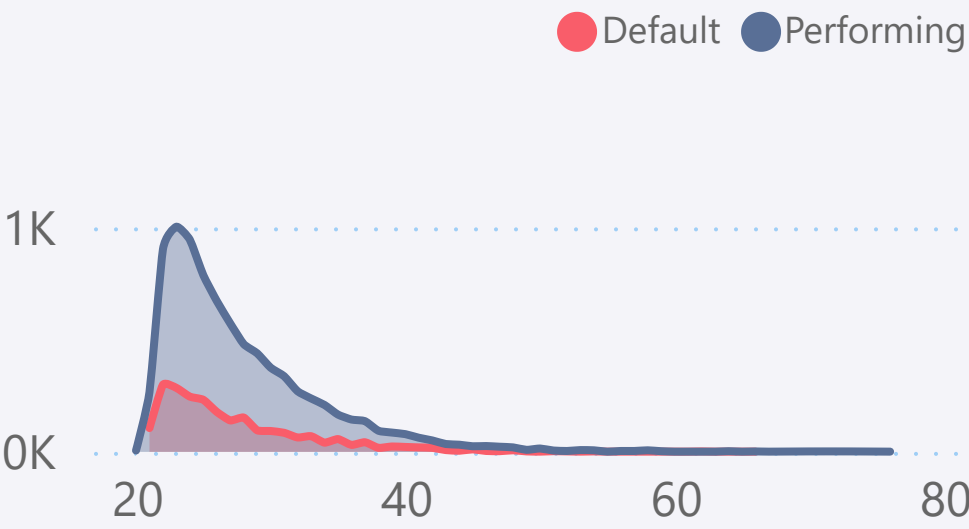


Portfolio at Risk
12.8M

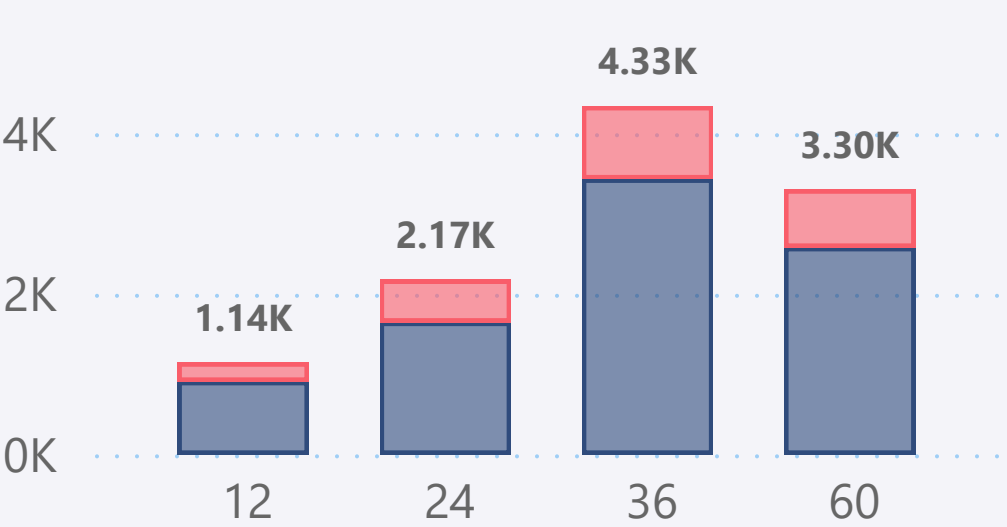
Loans by Amount



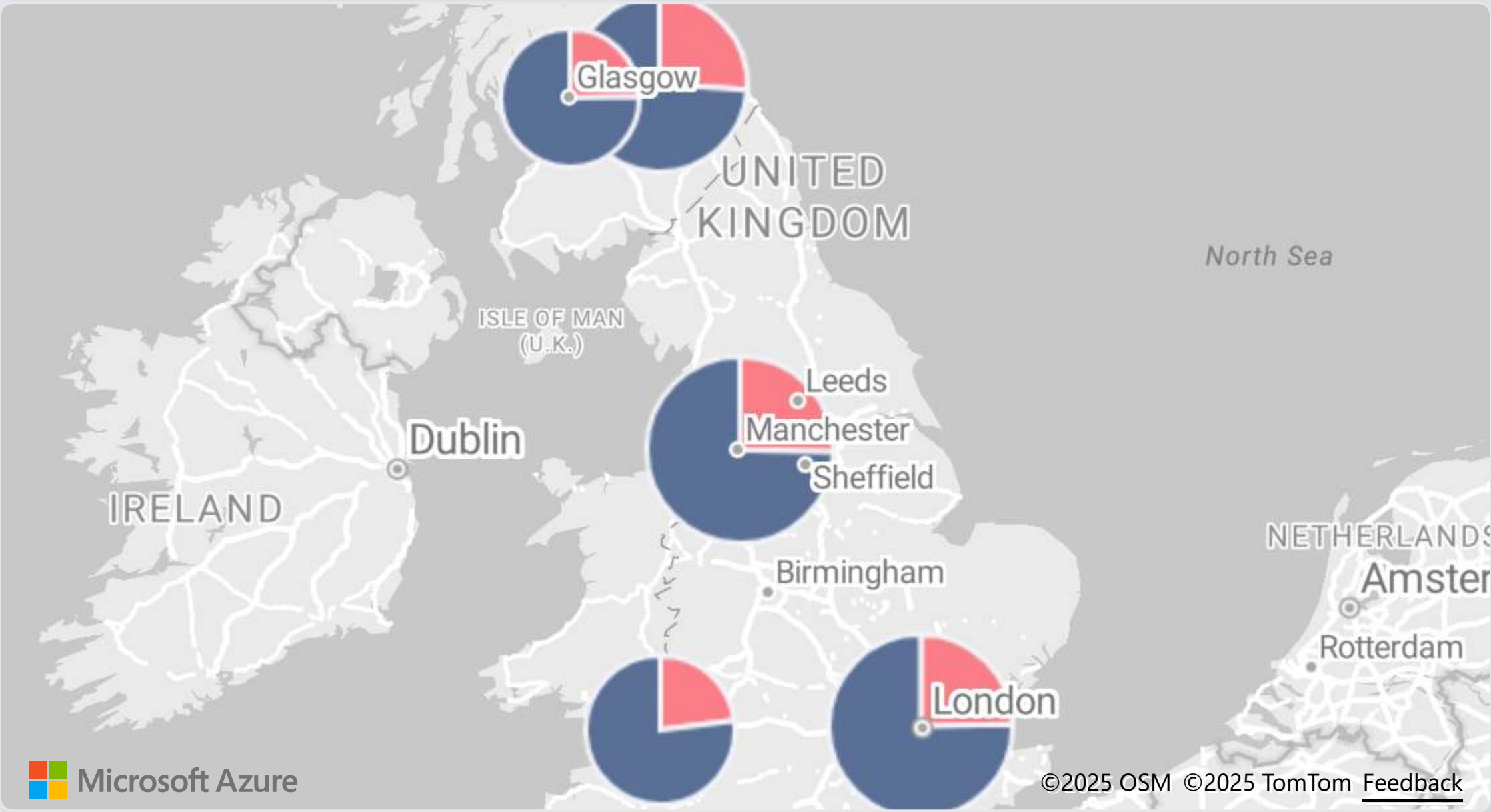
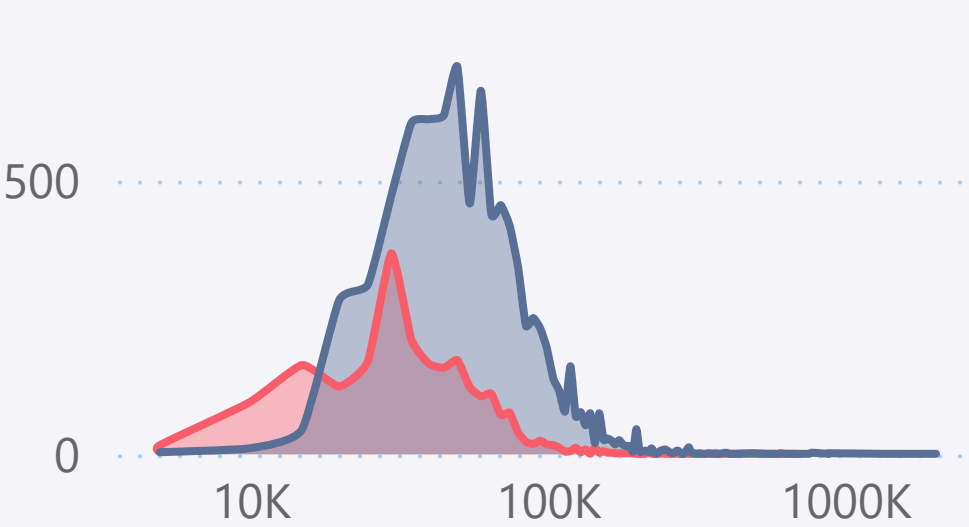
Borrowers Age



Loans by Term



Borrowers Income

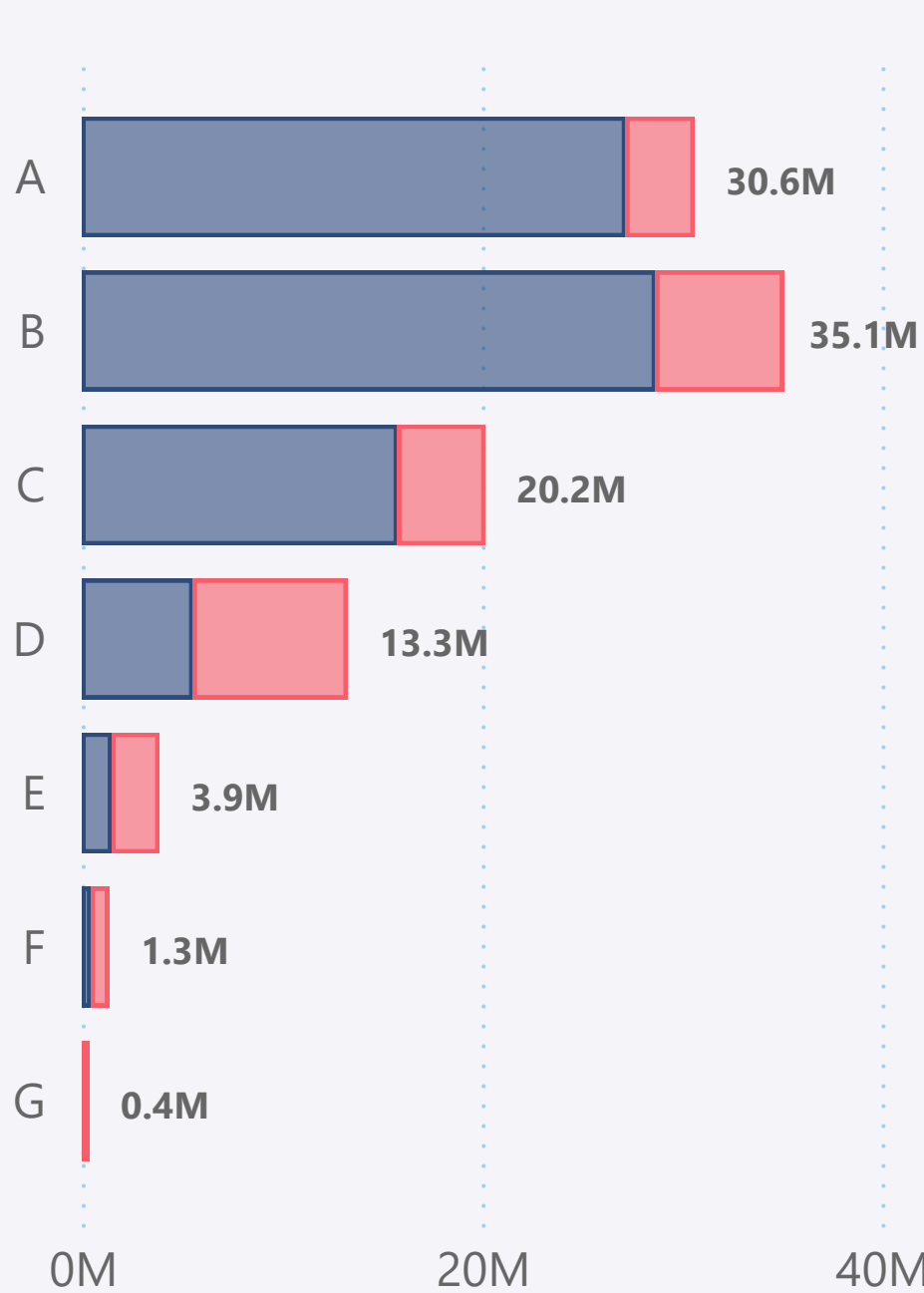


☐ Canada

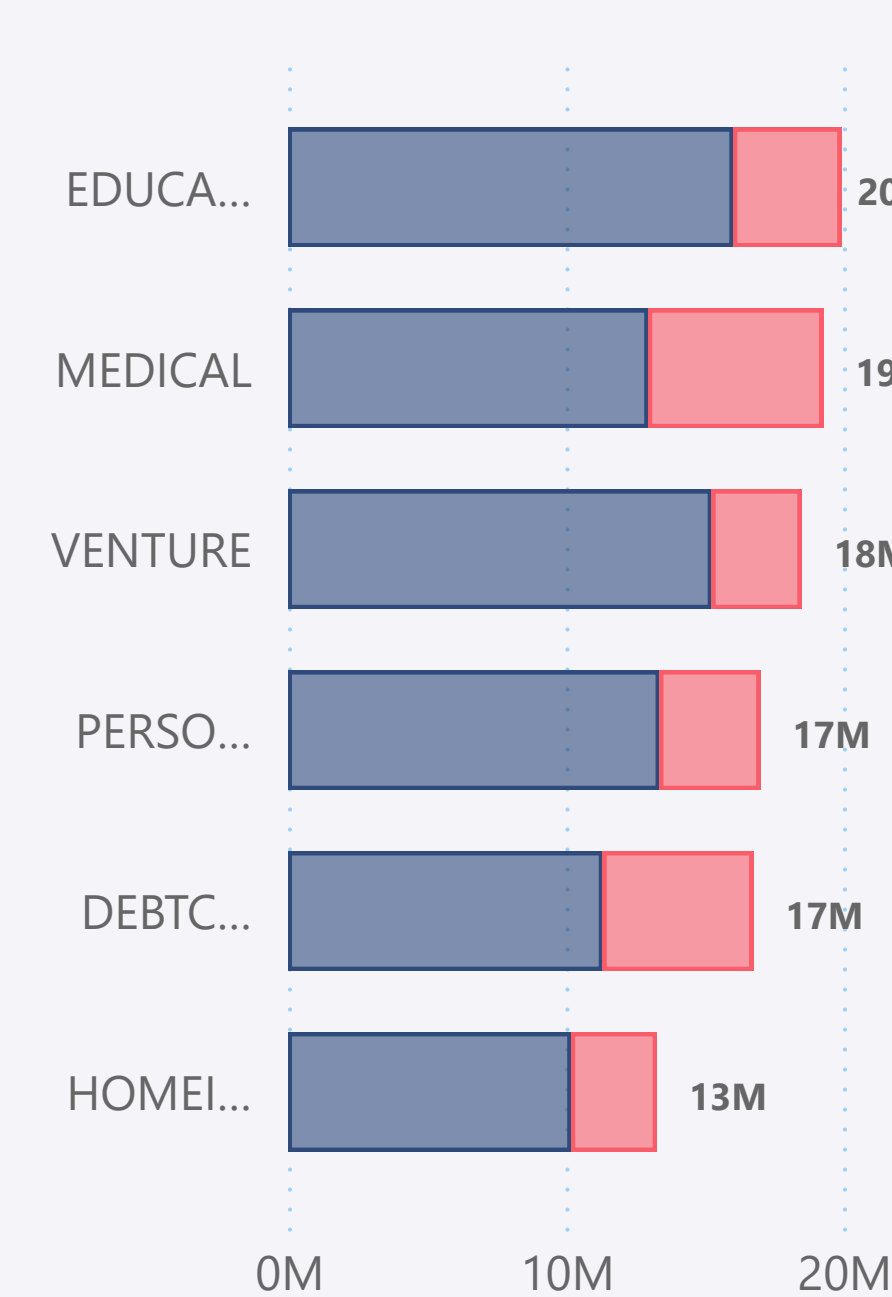
☒ UK

☐ USA

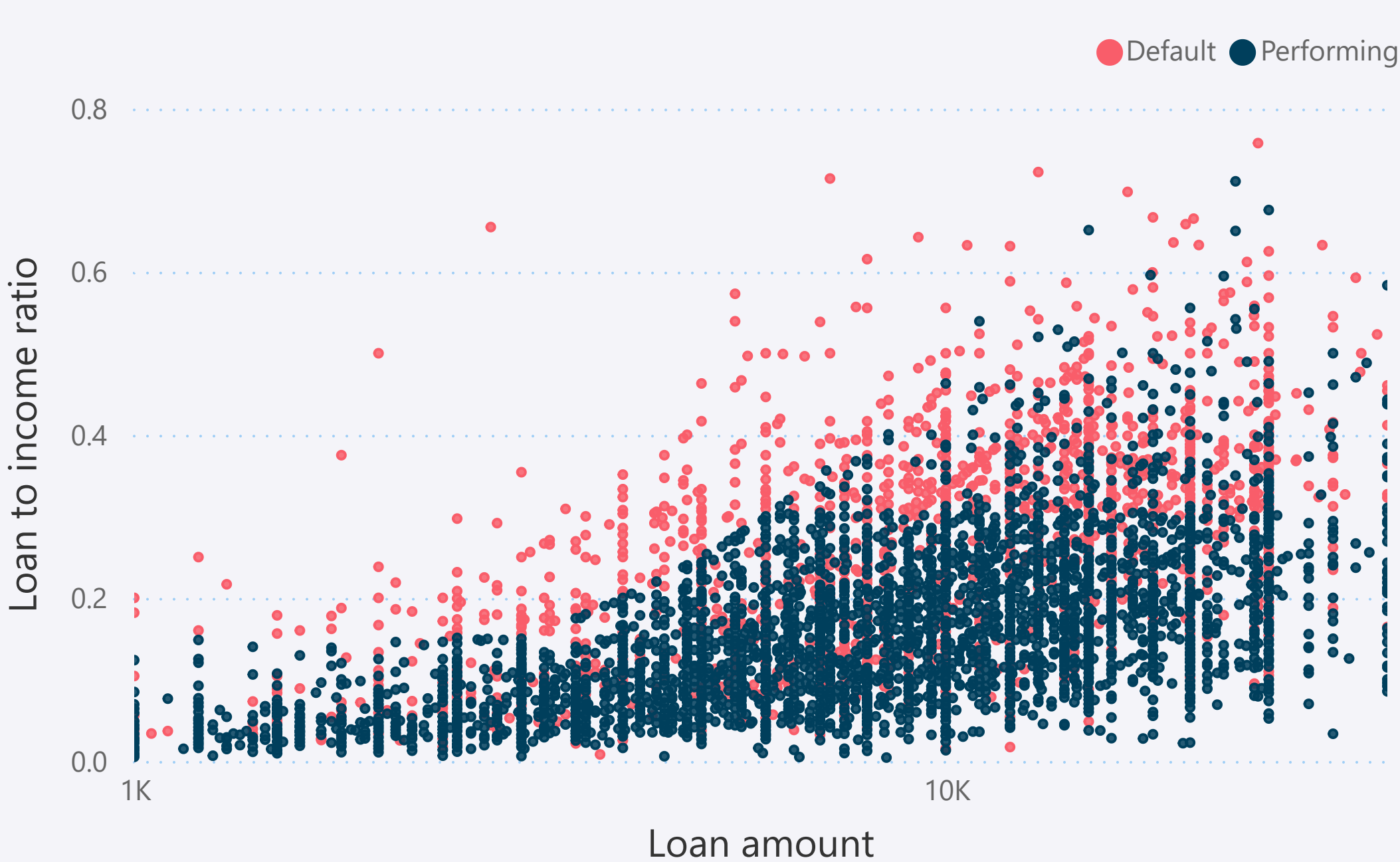
Loan Amount by Loan Grade



Loan Amount by Loan Intent



Loan Status by LTI and Loan Amount



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Precision
0.8494



Recall
0.9572

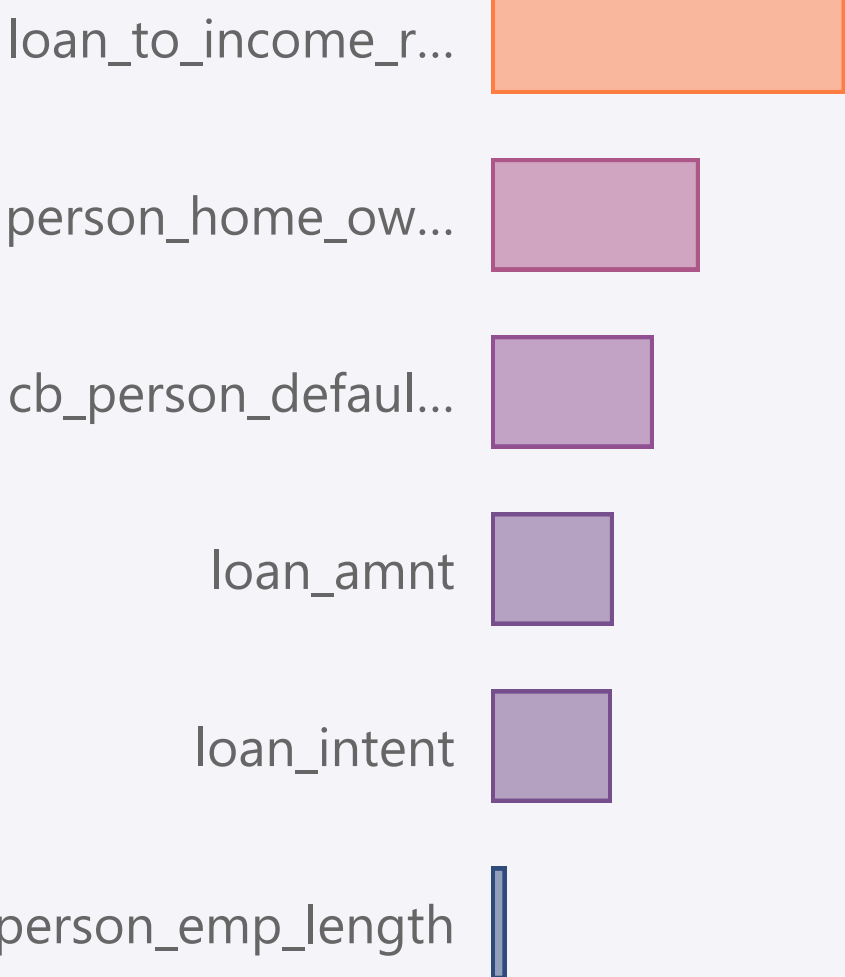


F1 score
0.9001



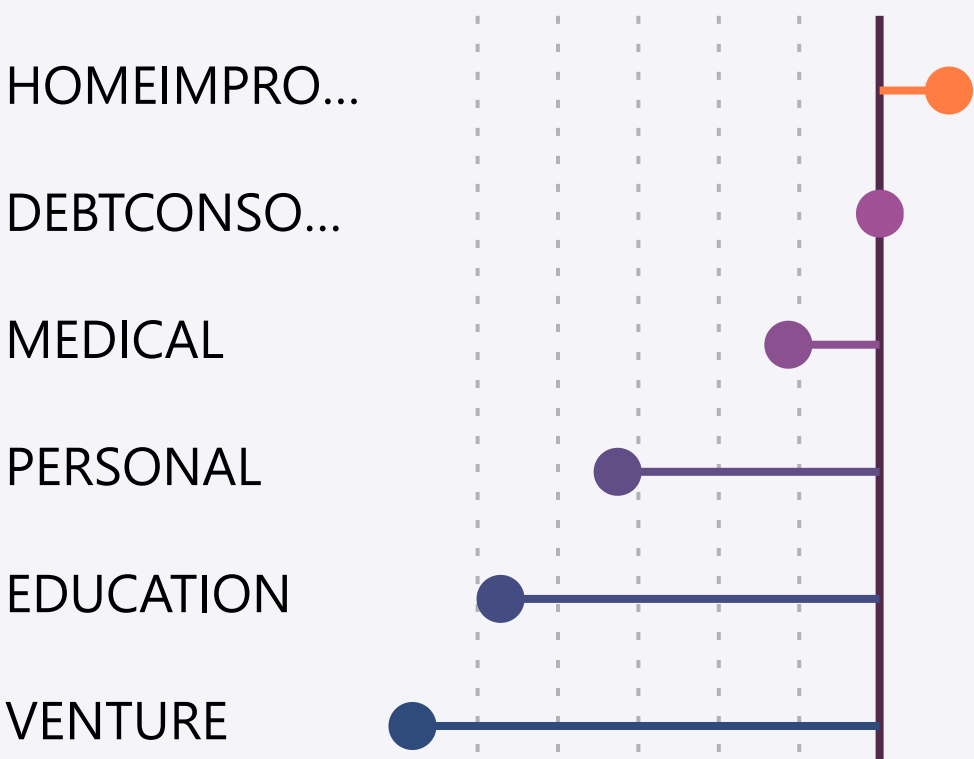
Balanced Accuracy
0.6798

Key factors

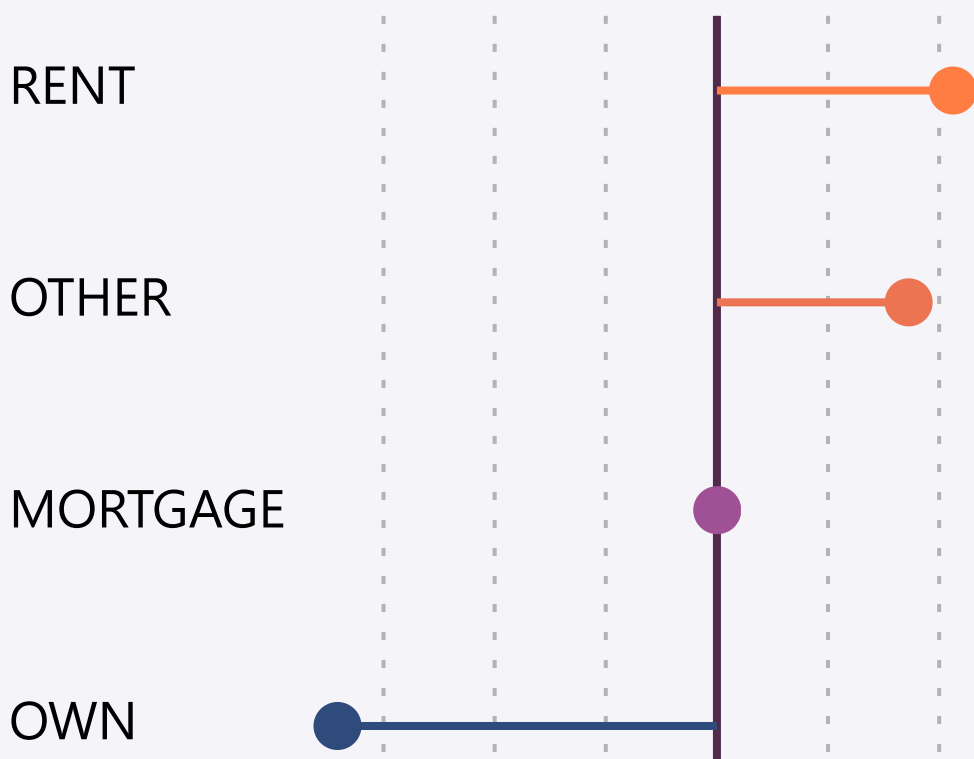


Marginal effects*

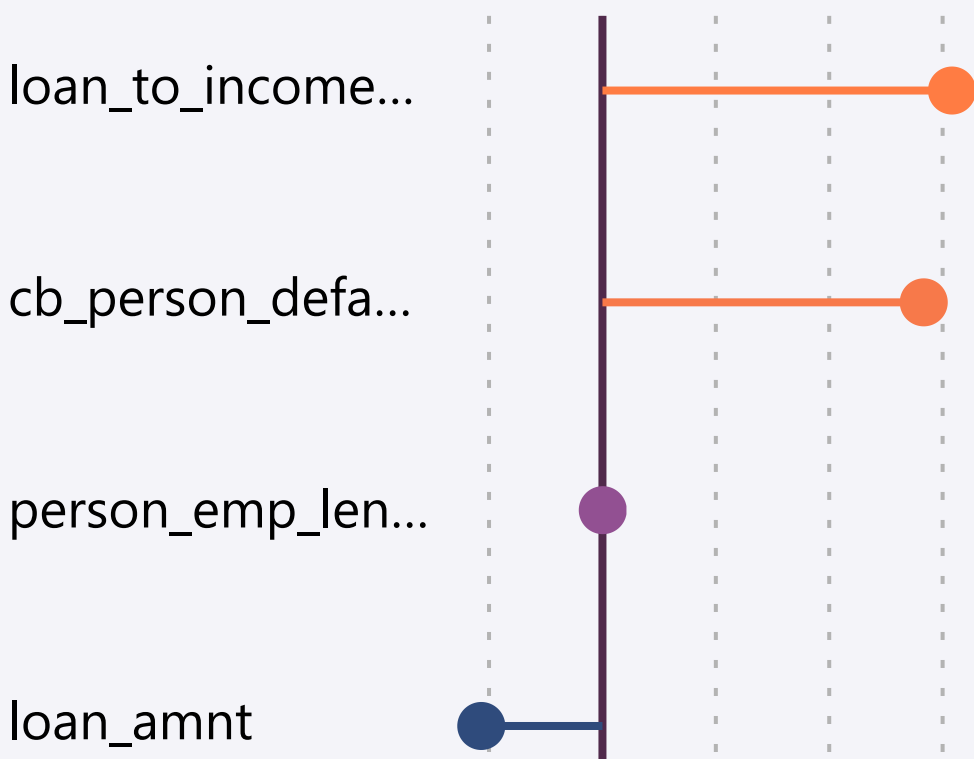
Loan intent



Home ownership



Other variables



* Standardized effects calculated using a Logistic Regression model

Default risk calculator*

Loan intent

- ☐ DEBTCONSOLIDATION
- ☐ EDUCATION
- ☐ HOMEIMPROVEMENT
- ☒ MEDICAL
- ☐ PERSONAL
- ☐ VENTURE

Home ownership

- ☐ MORTGAGE
- ☐ OTHER
- ☒ OWN
- ☐ RENT

Loan amount

20,400

Person income

67,000

CB Default on File

- ☐ N
- ☒ Y

Employment length

5

Predicted PD



* Calculated using a Logistic Regression model



Scenario Settings

Proportions

DEBTCONSOLIDATION

40

EDUCATION

50

HOMEIMPROVEMENT

41

MEDICAL

10

PERSONAL

75

VENTURE

50

Parameters

Borrowers Count

38,000

Bank Margin

32%

Recovery Rate

77%

Loan-to-income Cut Off

99%

Default-on-file Cut Off

0%

Rent and Other Cut Off

0%

Results Summary

Expected Portfolio Volume
363.8M

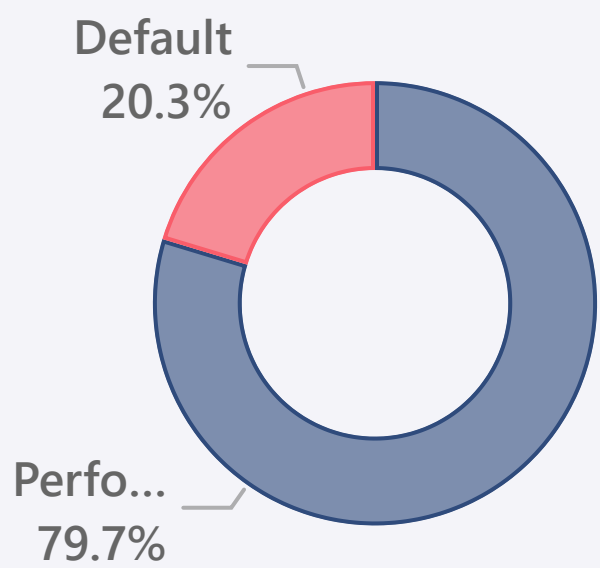
Expected Interest
41.1M

Expected Gross Margin
13.1M

Expected Default Rate
20.3%

Expected Loss
-9.2M

Expected Net Profit
3.9M



Simulations are conducted on a synthetic dataset generated by sampling the original data with added jitter. The probability of default (PD) is predicted using an XGBoost model trained on the original dataset. The "Default-on-file" and "Rent and Other" cutoffs exclude a specified percentage of borrowers with the highest predicted PD.

Metric	Value
Balanced Accuracy	0.8421
F1	0.9508
Precision	0.9205
Recall	0.9831
Sensitivity	0.9831
Specificity	0.7012

Loan Intent	Borrowers	Default R...	Loan Portfolio	Interest	Gross Margin	Non-recov. ...	Net Profit
DEBTCONSOLID...	5,485	28.6%	52.7M	5.9M	1.9M	-1.9M	0.0M
EDUCATION	8,488	16.7%	79.9M	9.0M	2.9M	-1.8M	1.1M
HOMEIMPROVE...	3,888	26.3%	40.4M	4.7M	1.5M	-1.4M	0.1M
MEDICAL	1,597	26.3%	14.9M	1.7M	0.5M	-0.5M	0.1M
PERSONAL	10,894	19.8%	103.8M	11.7M	3.8M	-2.4M	1.4M
VENTURE	7,523	14.7%	72.0M	8.0M	2.6M	-1.3M	1.2M
Total	37,875	20.3%	363.8M	41.1M	13.1M	-9.2M	3.9M

Calculated on a synthetic dataset with the original distributions